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TopBP1 coordinates DNA repair synthesis in mitosis via recruitment of the nuclease scaffold SLX4



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The majority of cancer cells experience replication stress, which ultimately causes them to enter mitosis with underreplicated DNA. To alleviate the consequences of replication stress, cells utilize a mechanism known as MiDAS that functions to complete synthesis of underreplicated DNA in early mitosis. This process is considered an Achilles heel for highly replicative cancers. In this study, we show that human TopBP1 localizes to sites of underreplicated DNA marked by FANCD2 and promotes MiDAS through recruitment of the nuclease scaffold protein SLX4. Additionally, we demonstrate that the recruitment of SLX4 to TopBP1 foci in mitosis depends on TopBP1-K704, SLX4-T1260, and several SUMO-interaction motifs in SLX4. Lastly, we show that the recruitment of SLX4 to TopBP1 foci in mitosis is important to prevent transmission of DNA damage to daughter cells. Based on this, we hypothesize that targeting the TopBP1-SLX4 interaction in mitosis may be a potential strategy for anti-cancer therapy.

Oncogene activation causes replication stress, which is thought to drive genome instability and tumorigenesis¹. Certain regions of the genome, called common fragile sites (CFSs), are particularly sensitive to replication stress as they contain DNA sequences that are difficult to replicate or have a scarcity of replication origins. CFSs often present as gaps and breaks on metaphase chromosomes, indicating the presence of underreplicated DNA². Additionally, CFSs typically reside within extremely large genes that may take more than one cell cycle to transcribe, thus making them prone to experience collisions between the transcription and replication machinery³. Although the bulk of DNA synthesis occurs in S phase, CFSs are prone to enter mitosis in an underreplicated state when challenged with replication stress⁴, which can compromise genome integrity when sister chromatids attempt to separate at the metaphase to anaphase transition. To avoid that underreplicated DNA leads to sister chromatid segregation defects in anaphase, cells utilize a specialized replication mechanism, called Mitotic DNA synthesis (MiDAS)^{5–7}. MiDAS resembles a break-induced replication (BIR) pathway where an incision at the stalled replication fork promotes conservative DNA synthesis⁸. Several proteins have been linked to the MiDAS pathway, including FANCD2, SLX4, RAD52, and POLD3. FANCD2 is

known to form “twin foci” at loci containing underreplicated DNA^{9–11} and is required for MiDAS^{12,13}, but the exact function of FANCD2 has not been established. The structure-selective nucleases XPF-ERCC1 and MUS81-EME1 are recruited to underreplicated DNA in mitosis^{14,15}. Importantly, scaffold protein SLX4 is required for the cleavage of stalled replication forks by nucleases such as MUS81¹⁶, which initiates MiDAS and recruitment of RAD52⁵. Notably, the majority of MiDAS in cancer cells requires RAD52 and not BRCA2/RAD51, suggesting strand annealing rather than strand invasion is utilized during MiDAS⁵. However, some MiDAS events also require RAD51¹⁷. MUS81 is recruited to SLX4 in a PLK1- or CDK1-dependent manner thereby limiting its activation to mitosis^{5,18,19}. Lastly, polymerase delta (POLD3), is recruited in an SLX4-dependent manner to promote DNA synthesis⁵.

Using chicken DT40 cells, we previously showed that TopBP1 is required for the recruitment of SLX4 to mitotic foci and that TopBP1 depletion impairs MiDAS²⁰. However, whether these functions of TopBP1 are conserved in humans and how TopBP1 recruits SLX4 in mitosis remains to be elucidated. In this study, we show that human TopBP1 localizes to sites of underreplicated DNA in mitosis and is required for MiDAS. The distinct

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localization pattern of TopBP1 relative to FANCD2 and MiDAS after replication stress, suggests that TopBP1 dissociates from chromatin after MiDAS is completed. Further, we dissect the interdependency of TopBP1 and SLX4 in mitosis. Specifically, we find that K704 of TopBP1 and T1260 of SLX4 likely mediate their mitotic phosphorylation-dependent interaction, leading to colocalization at mitotic foci. This colocalization also depends on the T1260-flanking SUMO-interacting motifs (SIMs) of SLX4. Finally, we show that the recruitment of SLX4 to mitotic TopBP1 foci is important for maintaining genome integrity following replication stress.

Results

TopBP1 localizes to sites of DNA synthesis in mitosis and promotes MiDAS

To assess whether TopBP1 is involved in MiDAS in human cells, we examined the localization of TopBP1 relative to known MiDAS-associated proteins on metaphase DNA, including FANCD2 and SLX4. To do so, we utilized a human HeLa cell line expressing GFP-tagged CAPH (Condensin 1) to visualize chromatin²¹ and Halo-tagged TopBP1 expressed from the endogenous loci. Cells were challenged for 16 h with mild replication stress by the polymerase inhibitor aphidicolin (APH). During the APH treatment, cells were synchronized at the G₂/M border by treatment with a CDK1 inhibitor (RO-3306). After synchronization, cells were released into mitosis and pulse-labelled with EdU for 35 min to visualize DNA synthesized in mitosis. Fluorescence microscopy revealed that human TopBP1 colocalizes with both FANCD2 and SLX4 at EdU positive loci on metaphase chromatin (Fig. 1A, B), which indicates that human TopBP1 is involved in the MiDAS pathway. To obtain further support for this conclusion, we utilized a human RPE1 cell line in which TopBP1 is endogenously tagged with a mAID/SMASH-double-degron-tag (mAS) on both alleles, which combines the small molecule-assisted shutoff (SMASH) technology²² with the mini-AID degron²³. Additionally, the cell line is induced by doxycycline to express OsTIR, which is an essential component of the AID-degron system. This cell line allows for conditional degradation of endogenous TopBP1 protein within 6 h (Fig. 1C and Supplementary Fig. 1). Thus, six hours before release into mitosis, TopBP1 was degraded by activation of the mAS-tag by treatment with Auxin (IAA) and Asunaprevir (ASV). Again, we synchronized cells in late G₂ with RO-3306 under mild replication stress. Following TopBP1 degradation and synchronization, cells were released into mitosis and pulse-labelled with EdU for 35 min. Mitotic cells were collected by mitotic shake-off and prepared for immunostaining. The lack of EdU incorporation after degradation of TopBP1 indicates that human TopBP1 is required for MiDAS (Fig. 1D). By contrast, FANCD2 foci in mitotic cells were unaffected by depletion of TopBP1 (Fig. 1E). Similar suppression of MiDAS was observed when degrading endogenous Halo-tagged TopBP1 using HaloProtac-E²⁴ in a HeLa cell line (Supplementary Fig. 2). Collectively, we find that human TopBP1 colocalizes with FANCD2 and SLX4 at sites of underreplicated DNA and promotes DNA synthesis at these loci in early mitosis.

TopBP1 displays asymmetric localization on metaphase DNA

Accumulation of underreplicated DNA after low-dose aphidicolin treatment was accompanied by a significant increase in the number of TopBP1 twin foci in early mitosis (Fig. 2A). Likewise, FANCD2, which is a marker of underreplicated DNA in mitosis^{2–11}, forms twin foci that often colocalize with TopBP1 (Figs. 1A and 2B). The colocalizing TopBP1 and FANCD2 foci were observed at telomeres, chromatin gaps, between sister chromatids and overlapping with chromatin (Fig. 2C). However, we were surprised to find that contrary to the TopBP1 twin foci seen in early mitosis (Fig. 2A, B), we frequently observed an asymmetric localization of TopBP1 foci relative to the symmetric FANCD2 foci at metaphase (Fig. 2C, D). Notably, we find that 65% of TopBP1 foci appear asymmetric relative to FANCD2 on metaphase sister chromatids (Fig. 2E). By contrast, 94% of FANCD2 foci are symmetric twin foci in metaphase. In summary, replication stress induces the formation of TopBP1 twin foci that colocalize with FANCD2 twin foci in early mitosis. While FANCD2 persists as twin foci, TopBP1 foci frequently

become asymmetric at the later stages of mitosis with TopBP1 localizing predominantly to one of the FANCD2 twin foci. Collectively, the data suggest that some TopBP1-bound structures are resolved in an asymmetric manner as cells progress through mitosis.

TopBP1 foci are disassembled upon completion of DNA synthesis in late mitosis

The data presented above suggest that TopBP1 is initially recruited to both sister chromatids at underreplicated loci in prometaphase. Hereafter, as cells progress to metaphase, one of the TopBP1 twin foci is disassembled, thus giving rise to the asymmetric localization of TopBP1 while FANCD2 remains symmetric. To test if the disassembly of some TopBP1 foci reflects the completion of DNA replication by MiDAS, we first examined the localization pattern of TopBP1 and FANCD2 relative to MiDAS as monitored by EdU incorporation on mitotic chromosomes. This analysis revealed 10 localization patterns (Fig. 3A) of which eight, representing 71% of foci, displayed an asymmetric localization of TopBP1 to one sister chromatid (A1–A8), while FANCD2 and EdU staining was observed to roughly the same extent on both sister chromatids (Fig. 3B–D). Additionally, we observed two symmetric patterns (S1–S2). The majority (80%) of the symmetric structures were characterized by colocalization of TopBP1, FANCD2, and EdU on both sister chromatids (S1). Assuming that MiDAS reflects conservative DNA replication⁵, the incorporation of EdU on both sister chromatids indicate that BIR is initiated on both sister chromatids.

To examine if the asymmetry of TopBP1 foci arises as a consequence of MiDAS, we examined TopBP1 localization relative to EdU under conditions that block MiDAS. Specifically, we released cells from a chemical G₂-block (RO-3306), which followed mild replication stress during the last S phase, into mitosis in the presence of a high-dose of APH (2 μ M) (Fig. 4A), which completely blocks polymerase activity and thus should inhibit EdU incorporation at underreplicated DNA⁵. As expected, a high-dose of APH during EdU-pulsing abolishes EdU incorporation in mitosis (Fig. 4B). However, we did not observe a block in the recruitment of TopBP1 (Fig. 4C), which indicates that TopBP1 recruitment is independent of MiDAS. Interestingly, we find that chemical inhibition of DNA polymerases in mitosis results in a concomitant decrease in asymmetric TopBP1 foci (Fig. 4D), indicating that active replication in mitosis is required for transforming TopBP1 twin foci into single foci. The asymmetry of TopBP1 at twin foci was also decreased when cells were released into mitosis in the presence of a RAD52 inhibitor (AICAR), which inhibits MiDAS⁵ (Supplementary Fig. 3).

Collectively, we find that TopBP1 is recruited to mitotic foci independently of MiDAS, while the asymmetric localization of TopBP1 at later stages of mitosis arises as a consequence of MiDAS.

MiDAS does not establish proper chromosome condensation to underreplicated loci

The asymmetric patterns of TopBP1 localization often coincided with a small or large chromatin gap on one of the sister chromatids (Fig. 3D). Moreover, initiation of RAD52-dependent MiDAS is known to rely on the SLX4 nuclease scaffold^{5,9}, which together with its associated nuclease, MUS81, is known to cause the “break-phenotype” on metaphase chromosomes after replication stress^{5,14,15}. The chromatin gaps were typically associated with the sister chromatid that contained the TopBP1 focus and they were typically EdU positive (Fig. 4E, F), which could suggest that dissociation of TopBP1 from chromatin is accompanied by its condensation after DNA replication or repair is completed. Since replication stress almost exclusively induced TopBP1 localization to EdU positive chromatin gaps on metaphase chromosomes (Fig. 4E, F), we examined whether DNA synthesis in mitosis is required for closing of gaps. However, blocking of MiDAS by inhibition of the DNA polymerase did not increase the number of TopBP1-bound condensin gaps (Fig. 4G), indicating that DNA synthesized by MiDAS fail to condensate²⁵, which may reflect the conservative mode of replication or inhibition of chromosome condensation by the DNA damage checkpoint²⁶.

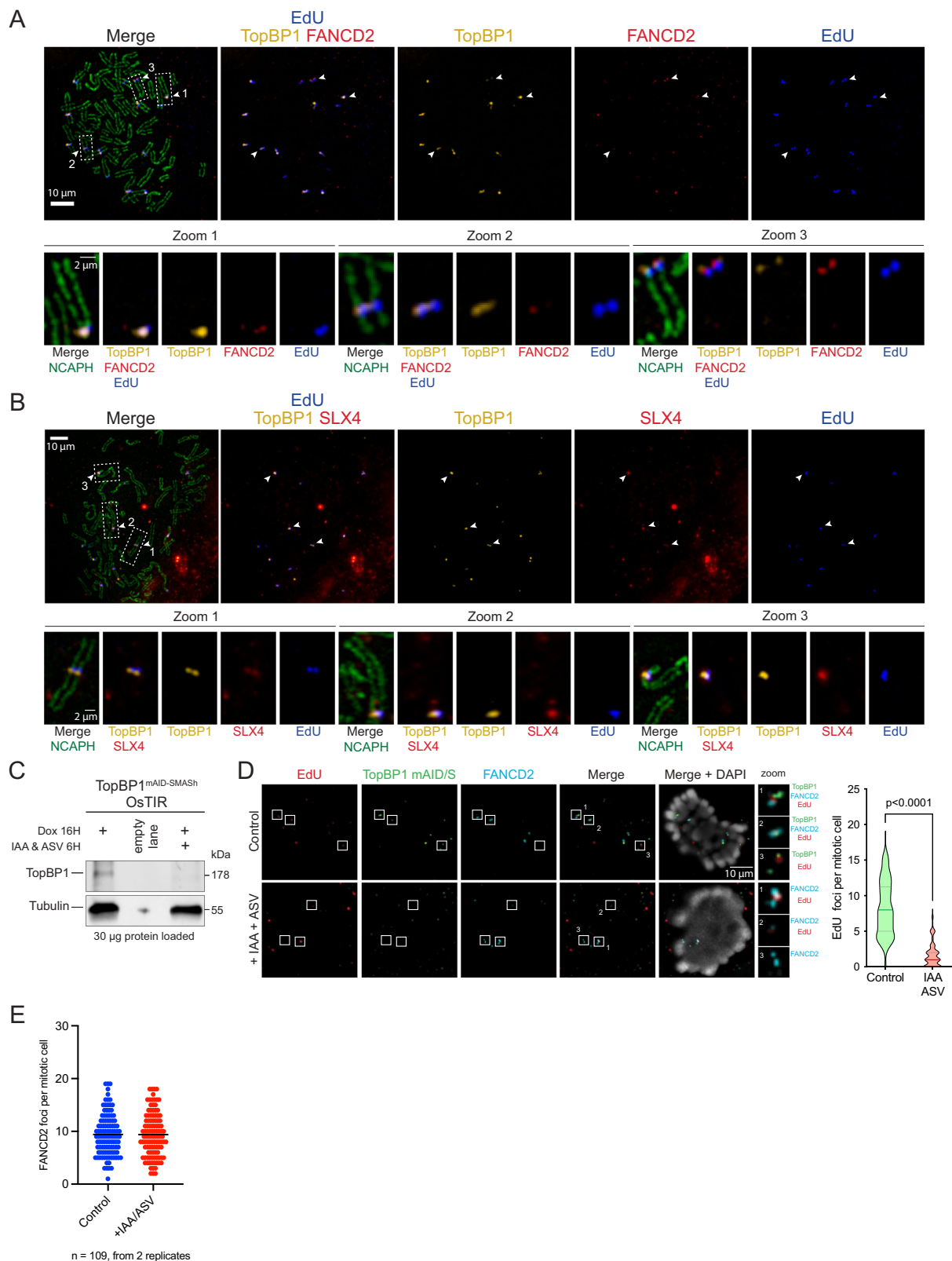
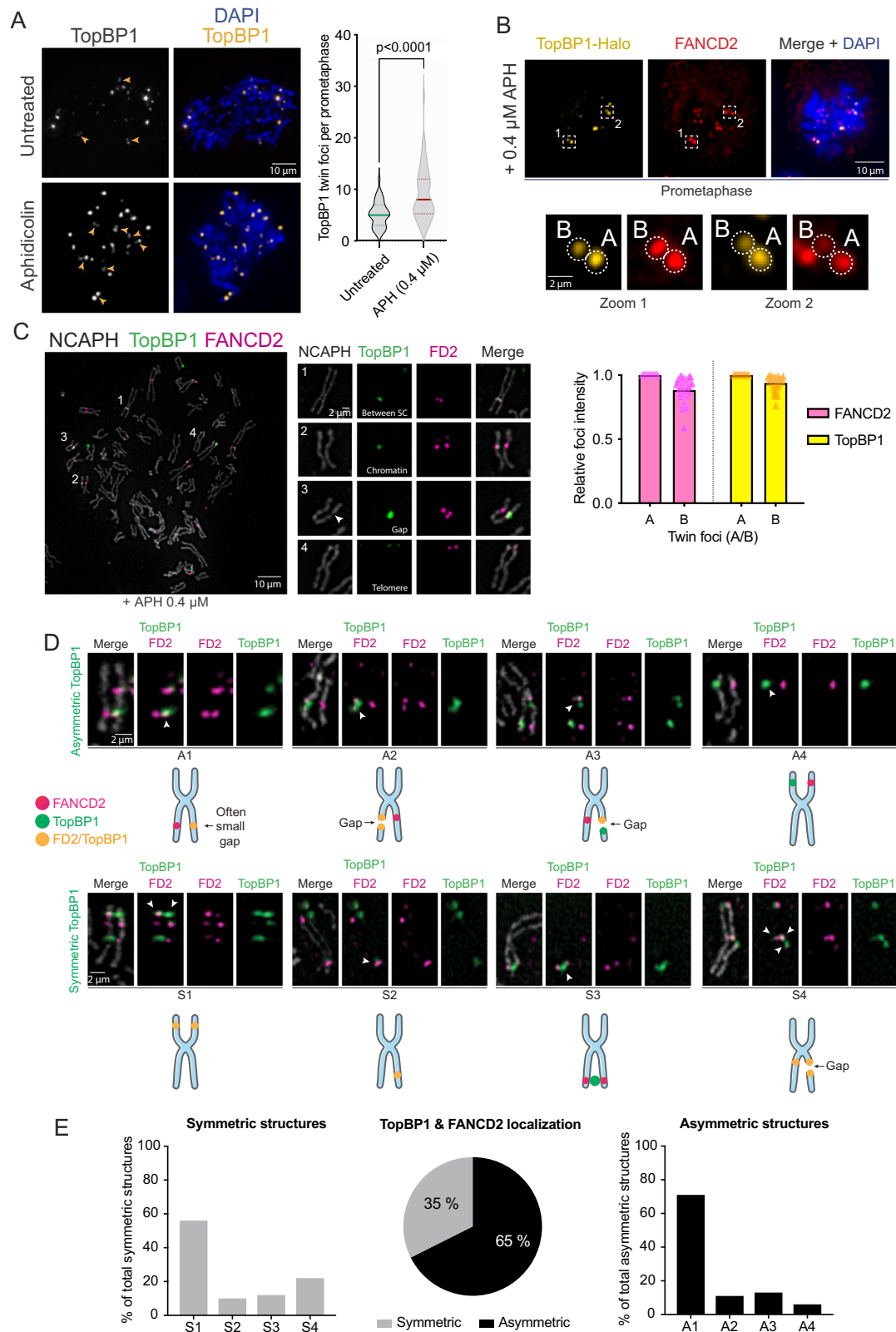


Fig. 1 | TopBP1 localizes to sites of mitotic DNA synthesis and promotes MiDas. **A, B** Representative images of native metaphase spreads showing the colocalization of TopBP1 and EdU with FANCD2 or SLX4 on CAPH-marked chromatin after mild replication stress (0.4 μ M APH, 16 h) in HeLa Kyoto cell line (NCAPH-mEGFP, TopBP1-Halo). CDK1i (RO-3306) was used for G2-synchronization prior to mitotic release. White arrows indicate foci on sister chromatids in boxed zoom. **C** Western blot showing degradation of endogenous TopBP1 after activation of mAID-SMASH

double-degron tag in RPE1 cells. **D** Representative images and quantification of TopBP1, EdU, and FANCD2 foci in prometaphase cells. Statistical significance was evaluated by Student's *t*-test ($n = 66$ (control) or 100 (+IAA)). Horizontal lines in violin plots indicate median and dotted lines indicate quartiles. CDK1i (RO-3306) was used for G2-synchronization prior to mitotic release. **E** Quantification of FANCD2 foci in the experiment described in panel D.



Taken together, we conclude that MiDAS does not establish proper chromosome condensation to underreplicated loci.

TopBP1-K704 is required for SLX4 recruitment to foci in mitosis

In chicken DT40 cells, TopBP1 directs the recruitment of SLX4 to foci during mitosis²⁰. Moreover, human TopBP1 was reported to interact with

SLX4²⁷. This interaction is controlled by CDK1-dependent phosphorylation of SLX4-T1260, a mode which is conserved in budding yeast²⁷. In yeast, the interaction between Slx4 and Dpb11/TopBP1 involves the BRCT 3–4 domains of Dpb11 (Fig. 5A)²⁷. The SLX4-interaction domain of human TopBP1 has not been identified. However, BRCT3–4 of Dpb11 in *Saccharomyces cerevisiae* resembles BRCT4–5 of human TopBP1²⁸.

Fig. 2 | TopBP1 and FANCD2 display asymmetric colocalization pattern on metaphase DNA. **A** Representative images and quantification of TopBP1 twin foci per prometaphase (chromosomes condensed but not yet aligned along the metaphase plate) cell after untreated or replication stress conditions in HeLa Kyoto cell line (NCAPH-mEGFP, TopBP1-Halo). Orange arrows indicate TopBP1 twin foci. Statistical significance was evaluated by Student's *t*-test ($n = 48$ (control) or 52 (APH)). Horizontal lines in violin plots indicate median and dotted lines indicate quartiles. Notably, CDK1-inhibition was not used in this experiment. **B** Representative images of TopBP1 and FANCD2 twin foci in early mitosis (prophase/prometaphase). Dashed squares indicate two zoom regions containing colocalizing twin foci of TopBP1 and FANCD2 on condensed DNA (DAPI). Quantification of relative TopBP1 and FANCD2 focus intensities are shown in bar plot. A indicates the sister chromatid containing the most intense TopBP1 focus, while B indicates the sister chromatid with the less intense TopBP1 focus. Each data point represents quantification from one focus ($n = 37$, 2 biological replicates). Notably, CDK1-inhibition was not used in this experiment. **C** Representative image

of a native metaphase spread showing the localization of TopBP1 and FANCD2 on chromatin (visualized by CAPH). Numbers indicate sister chromatids shown in zoom view. 1–4 indicates four different types of TopBP1 and FANCD2 localization on chromatin (between sister chromatids, on chromatin, in gaps, and on telomeres) after mild replication stress ($0.4 \mu\text{M}$ APH, 16 h). White arrow indicates a gap on the chromatin. **D** Representative images and illustrations of TopBP1-FANCD2 localization patterns on metaphase chromatin. A1–A4 illustrate asymmetric localization patterns between TopBP1 and FANCD2. S1–S4 illustrate symmetric localization patterns between TopBP1 and FANCD2. White arrows indicate the TopBP1-FANCD2 localization pattern illustrated below images. **E** Quantification of TopBP1-FANCD2 symmetry from data presented in (C). Pie-chart shows the percentages of symmetric/asymmetric patterns of TopBP1-FANCD2 foci on metaphase chromatin. Bar-plots show the percentage contribution of the different symmetric (S1–S4) and asymmetric (A1–A4) TopBP1-FANCD2 structures ($n = 458$). C–E CDK1i (RO-3306) was used for G2-synchronization prior to mitotic release.

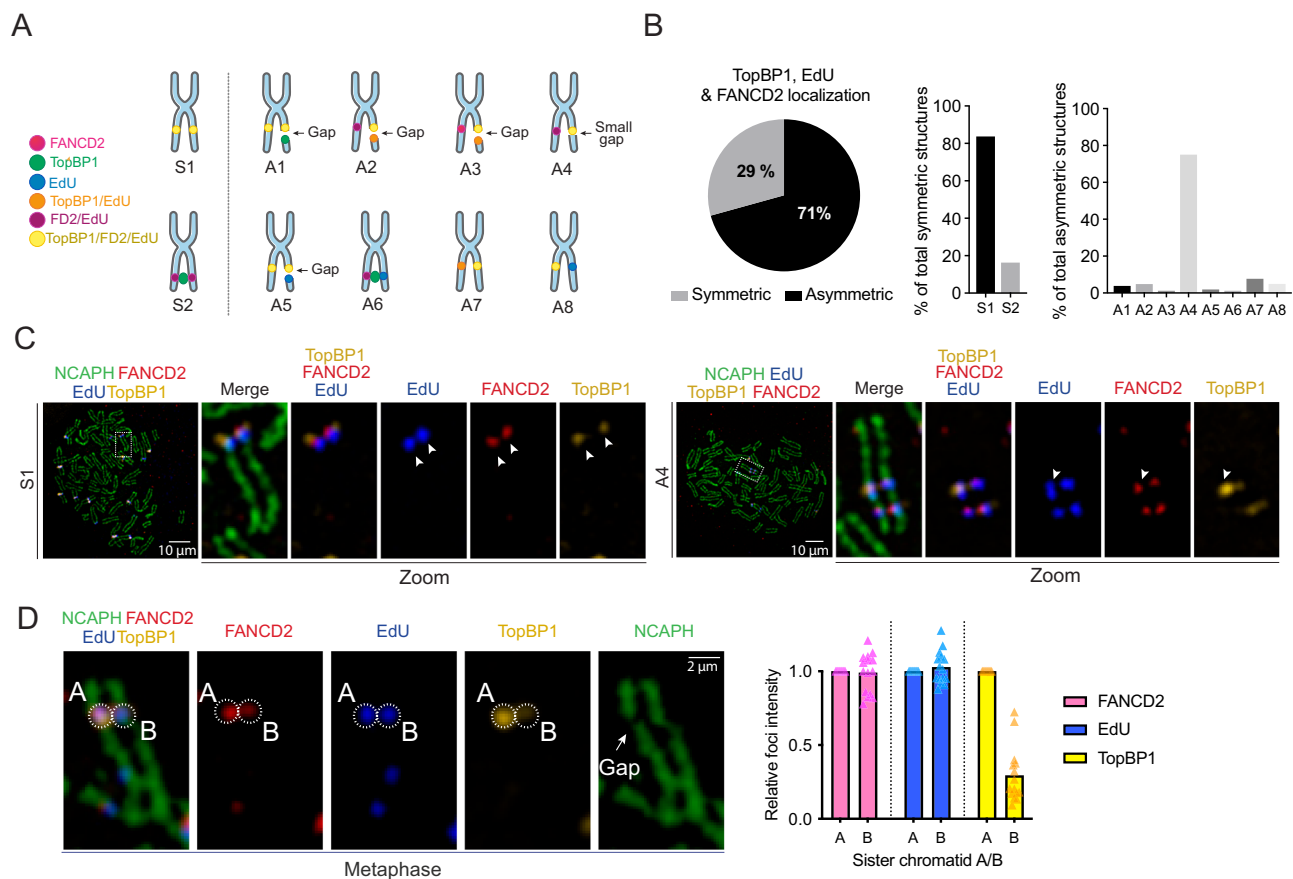


Fig. 3 | TopBP1 localizes asymmetrically to chromatin gaps relative to FANCD2-EdU twin foci on metaphase chromosomes after replication stress. **A** Illustrations of TopBP1-FANCD2-EdU localization patterns on metaphase chromatin in the HeLa Kyoto cell line (NCAPH-mEGFP, TopBP1-Halo). A1–A8 illustrate asymmetric localization patterns between TopBP1, FANCD2, and EdU. S1–S2 illustrate symmetric localization patterns between TopBP1, FANCD2, and EdU. **B** Quantification of TopBP1-FANCD2-EdU symmetry. Pie-chart shows the percentages of symmetric/asymmetric patterns of TopBP1-FANCD2-EdU foci on metaphase chromatin. Bar-plots show the percentage contribution of the different symmetric (S1–S2) and asymmetric (A1–A8) TopBP1-FANCD2-EdU structures ($n = 163$). **C** Representative images of the two most highly represented symmetric/

asymmetric localization patterns of TopBP1, FANCD2, and EdU. White box indicates area of zoom. White arrow indicates foci that colocalize with TopBP1. **D** Representative image of asymmetric structure A4. Dashed circles indicate areas of quantification, where intensities of TopBP1, FANCD2, and EdU were measured. A indicates the sister chromatid containing the most intense TopBP1 focus, while B indicates the sister chromatid lacking (or with very dim) TopBP1 focus. Quantification of relative TopBP1, FANCD2, and EdU focus intensities are shown in bar plot. Each triangle indicates quantification from an A4 asymmetric structure ($n = 15$, 2 biological replicates). B–D CDK1i (RO-3306) was used for G2-synchronization prior to mitotic release.

Moreover, BRCT5-K704 is known to mediate interactions with other DNA repair factors such as BLM²⁹, 53BP1³⁰, and BRCA1³¹, and this residue was reported to promote the recruitment of TopBP1 to ultrafine anaphase bridges³². Thus, to examine if the Slx4-interacting domain of Dpb11 is conserved in human cells, we constructed cell lines with

doxycycline-inducible expression of mCherry-tagged wild-type (WT) or mutant TopBP1 using the Flp-In system³³ where the conserved K704 in BRCT5 of TopBP1 was mutated to alanine. In these cell lines, we also ectopically expressed mVenus-tagged SLX4 to examine the colocalization of SLX4 with TopBP1. An equal expression of both proteins was observed

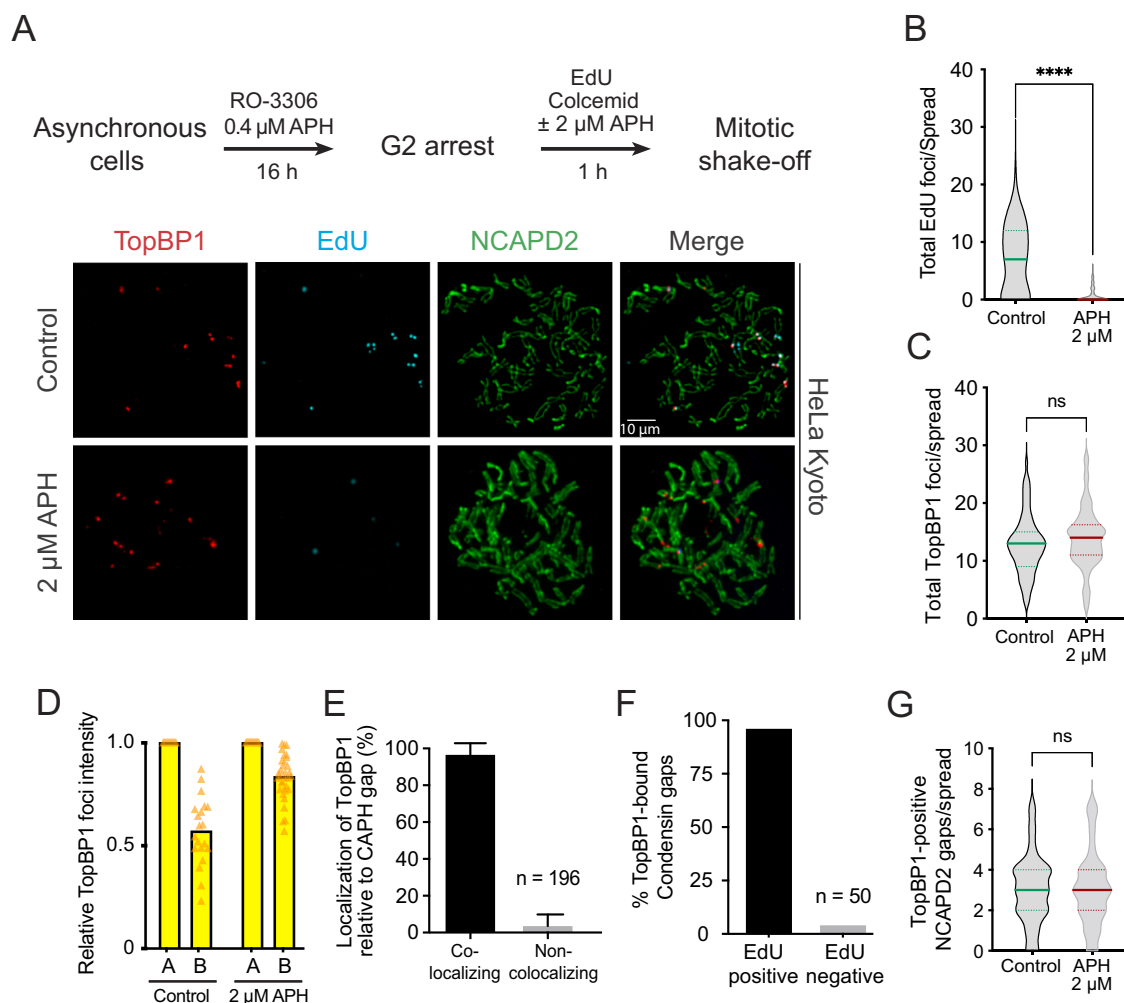


Fig. 4 | TopBP1 asymmetry is dependent on mitotic DNA replication.

A Experimental scheme and representative images of TopBP1 and EdU foci on CAPH-visualized metaphase chromosomes with or without high-dose APH (2 μM, 1 h) during EdU pulse in early mitosis in HeLa Kyoto cell line (NCAPH-mEGFP, TopBP1-Halo). **B** Quantification of total EdU foci per metaphase spread with or without high-dose APH (2 μM, 1 h). Statistical significance was evaluated by Student's *t*-test. ****, $p < 0.0001$ ($n = 77$). Horizontal lines in violin plots indicate median and dotted lines indicate quartiles. **C** Quantification of total TopBP1 foci per metaphase spread with or without high-dose APH (2 μM, 1 h). Statistical significance was evaluated by Student's *t*-test ($n = 77$). Horizontal lines in violin plots indicate median and dotted lines indicate quartiles. ns, not significant. **D** Quantification of relative intensity of TopBP1 in twin foci under inhibition of MiDAS with or without high-dose APH (2 μM, 1 h). A indicates the sister chromatid containing the brighter TopBP1 focus, while (B) indicates the sister chromatid with

the dimmer TopBP1 focus. Each triangle in the bar plot indicates quantification of a pair of TopBP1 twin foci from (A) ($n = 20$ (Control) or $n = 34$ (2 μM APH)). Two biological replicates. **E** Quantification of the localization of TopBP1 relative to chromatin gaps. "Colocalizing" indicates the percentage of chromatin gaps that contain a TopBP1 focus. "Non-colocalizing" indicates chromatin gaps devoid of TopBP1 focus. $n = 196$ indicates that 196 chromatin gaps were evaluated for colocalization with TopBP1. Error bars indicate 95% confidence intervals. **F** Quantification showing the percentage of TopBP1-bound chromatin gaps that are EdU positive or negative. $n = 50$ indicates that 50 TopBP1-bound chromosome gaps were evaluated for colocalization with EdU. **G** Quantification of TopBP1 positive chromatin gaps per metaphase spread with or without high-dose APH (2 μM, 1 h). Statistical significance was evaluated by Student's *t*-test ($n = 23$). Horizontal lines in violin plots indicate median and dotted lines indicate quartiles. ns, not significant. **B–G** CDK1i (RO-3306) was used for G2-synchronization prior to mitotic release.

by western blotting (Supplementary Fig. 4A). As expected, WT TopBP1 and SLX4 showed a high degree of colocalization in mitosis (Fig. 5B, C). In contrast, using the mutated TopBP1 (K704A), we found that colocalizing SLX4 foci were barely detectable (Fig. 5B–D), while TopBP1-K704A was fully proficient in focus formation. Collectively, this suggests that the TopBP1-dependent recruitment of SLX4 to foci in mitosis is largely dependent on K704 in the BRCT5 of TopBP1.

SLX4-T1260 and SIMs are required for the association with TopBP1 in mitosis

The interaction between TopBP1 and SLX4 has previously been suggested to be regulated by two different mechanisms: (1) cell cycle-dependent phosphorylation by CDK1 and (2) DNA damage-induced PTMs²⁷. Notably, TopBP1 is one of the most phosphorylated and SUMOylated proteins in

response to replication stress³⁴, and SLX4 contains three SIMs that are in close proximity to T1260, which was reported to promote the binding to TopBP1²⁷ (Fig. 6A). Therefore, we hypothesized that SUMOylation of TopBP1 following replication stress may enhance or stabilize the interaction between TopBP1 and SLX4. To investigate this, we utilized the Flp-In system to introduce doxycycline-inducible expression of SLX4 cDNA (WT, T1260A, ΔSIM1-3, or ΔSIM1-2) (Fig. 6A) in a HeLa cell line ectopically expressing TFP-tagged H2B to visualize chromatin and Halo-tagged TopBP1 from the endogenous locus. Equal SLX4 protein expression was validated by western blotting (Supplementary Fig. 4B). The SLX4 ΔSIM1-3 or ΔSIM1-2 mutants contain alanine substitutions of key residues in the SIMs of SLX4 to abolish its interaction with SUMOylated proteins (Fig. 6A)³⁵. WT and mutant cells were treated with low-dose APH (0.4 μM) and doxycycline (1 μg/ml) for 16 h before labelling TopBP1 with the

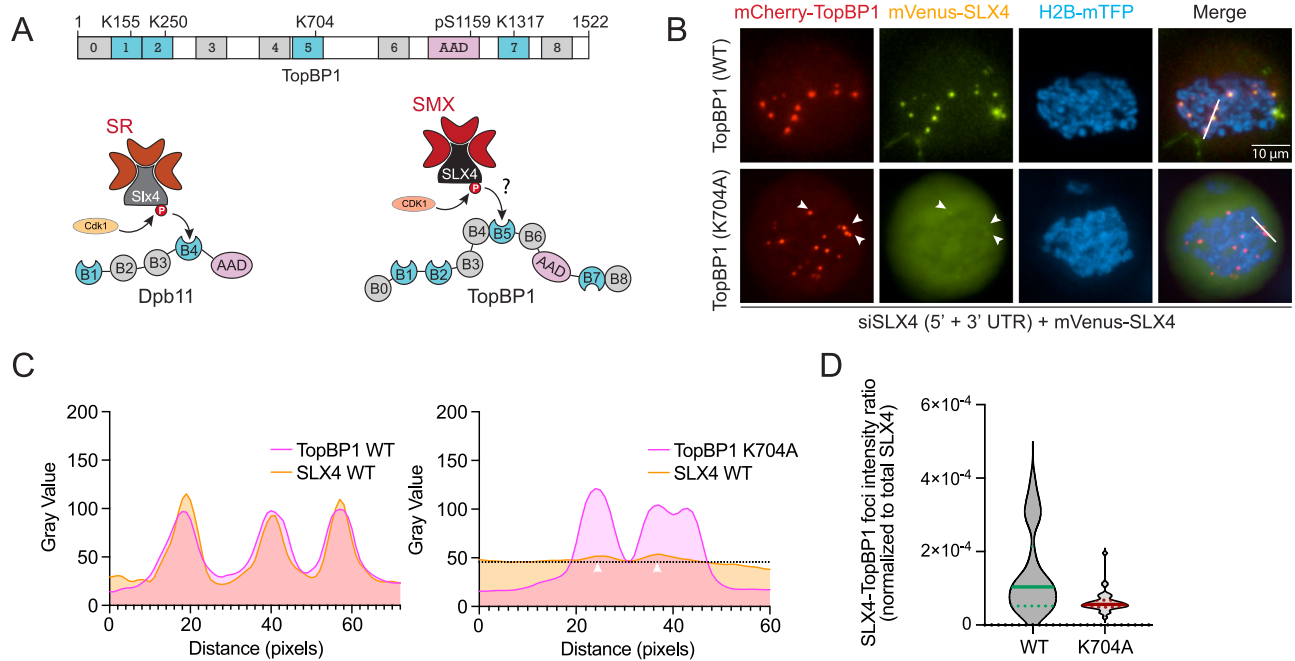


Fig. 5 | TopBP1 K704 is required for SLX4 recruitment in mitosis. **A** Illustration of human TopBP1 domain organization. Blue boxes indicate BRCT-domains capable of binding phosphorylated residues through an internal lysine residue. Schematic summary of Dpb11-Slx4 interaction in yeast²⁷ and hypothesis for interaction in human. SR (Slx1-Rad1). SMX (Slx1-MUS81-XPB). **B** Representative images of HeLa cells expressing mCherry-tagged WT or mutant (K704A) TopBP1 cDNA from FRT site, treated with APH (0.4 μ M) for 16 h before imaging. Cells were subjected to knockdown of SLX4 from endogenous locus with siRNA's targeting 5' and 3' UTR and complemented with Venus-tagged SLX4 cDNA by transient transfection. Notably, knockdown of endogenous SLX4 was performed twice. White arrows indicate colocalizing foci in TopBP1-K704A cells. White lines crossing

TopBP1-SLX4 foci indicate the line used for quantification of TopBP1-SLX4 intensity values (gray value) plotted in (C). Notably, CDK1-inhibition was not used in this experiment. **C** Plot showing the intensity values of TopBP1 (WT or K704A) and SLX4 (WT) along the white lines indicated in (B). White arrows indicate the two foci that are intersected by the white line from the K704A images in (B). Dashed black line indicates the background intensity of SLX4. **D** Quantification of colocalizing focus intensities. Plotted values indicate the intensity-ratio between SLX4 and TopBP1 at colocalizing foci, normalized to the total SLX4 signal per nucleus. Horizontal lines in violin plots indicate median and dotted lines indicate quartiles ($n = 34$ (WT) or $n = 37$ (K704A), 2 replicates).

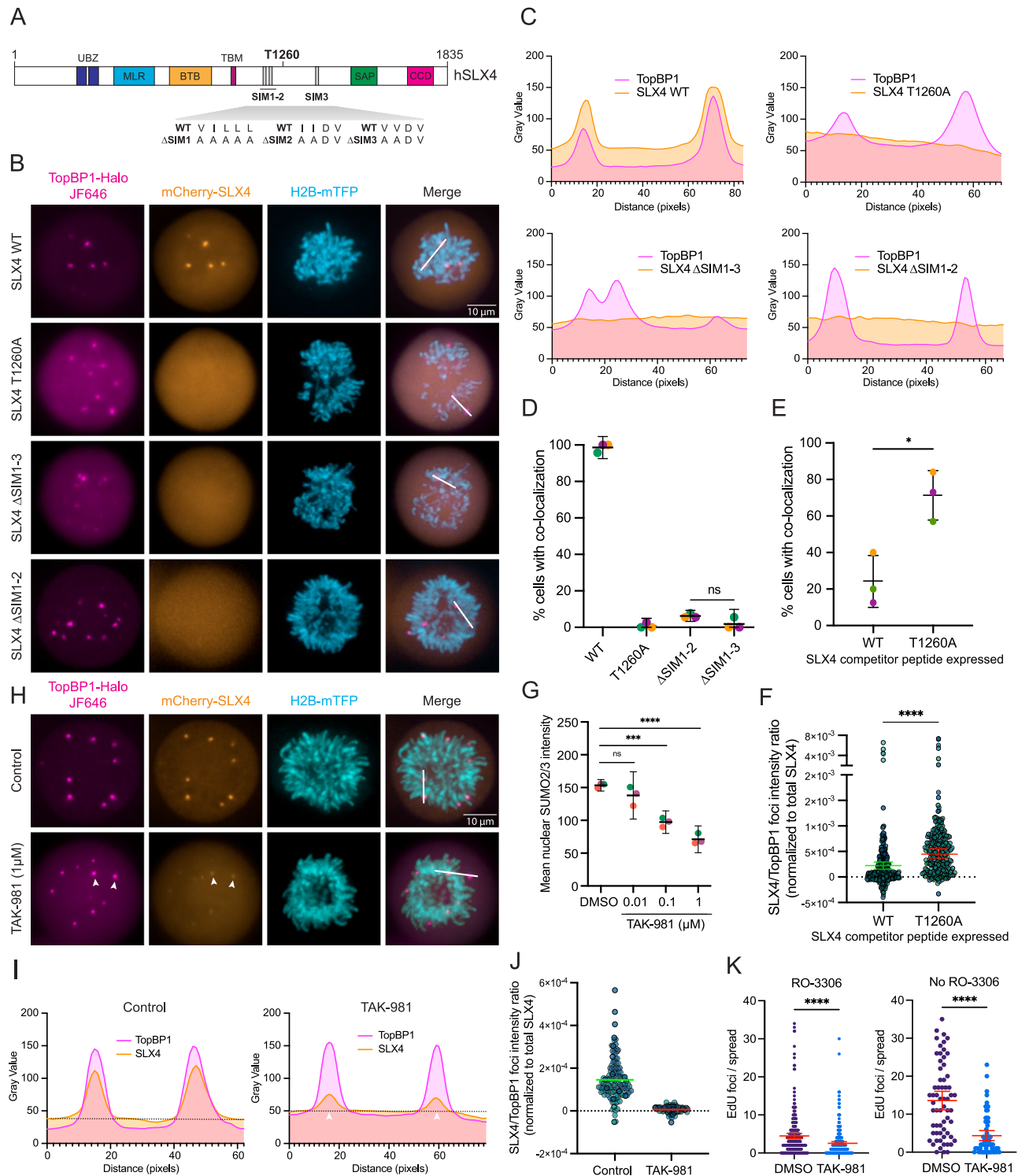
near-infrared Halo-ligand, JF646. In accordance with our previous observations (Fig. 5B), WT TopBP1 colocalized with WT SLX4 in mitosis (Fig. 6B–D). In contrast, mutating the CDK1-target residue, T1260^{27,36}, in SLX4 abolished SLX4 focus formation in mitosis (Fig. 6B–D). Interestingly, a similar phenotype was observed when mutating the two upstream SIMs (Δ SIM1-2) alone or together with the downstream SIM (Δ SIM1-3), even in the presence of T1260 (Fig. 6B–D). Of note, we do not find any difference in the localization of the Δ SIM1-2 and Δ SIM1-3 mutants of SLX4. Thus, the SIM1-2 motifs, which are in closest proximity to T1260, are likely the main contributor to this SIM-dependent colocalization of SLX4 with TopBP1. To further validate the interaction between TopBP1 and SLX4-T1260, we expressed competitor peptides of 17 amino acids representing the region around SLX4-T1260 in either the WT form or with the T1260A mutation to investigate if the WT peptide could compete with localization of mCherry-SLX4 to TopBP1 foci. The expression was controlled by fusing the peptide to a FKBP12-based destabilizing domain, which ensures continuous degradation of the peptide, unless the domain is stabilized with the Shield-1 ligand³⁷. Expression of the WT peptide significantly decreased the colocalization between TopBP1 and SLX4, when compared to the T1260A peptide (Fig. 6E, F and Supplementary Fig. 5), supporting that SLX4-T1260 plays a direct role in the TopBP1-SLX4 interaction. Collectively, these results suggest that both T1260 and SIMs are necessary for the recruitment of SLX4 to TopBP1 foci in mitosis.

To further substantiate the SUMO-dependency of the TopBP1-SLX4 colocalization, we examined if inhibition of SUMO-conjugation influences the recruitment of SLX4 to TopBP1 foci in mitosis. Dosage-dependent inhibition of SUMO-conjugation could be achieved by incubating cells with the SUMO-inhibitor, TAK-981, for 4 h (Fig. 6G)³⁸. After treatment with

TAK-981, we find that the relative intensity of TopBP1-colocalizing SLX4 foci was significantly reduced (Fig. 6H–J), which suggests that an active SUMOylation pathway contributes to recruiting SLX4 to TopBP1 foci in mitosis. To investigate the effect of TAK-981 on MiDAS, we examined the number of EdU foci on metaphase chromatin in cells exposed to mild replication stress for 16 h and TAK-981 for 2 h (Fig. 6K). The SUMO-inhibitor led to a decrease in the number of EdU foci, indicating that SUMO-dependent interactions are important for promoting MiDAS. Importantly, mitotic EdU incorporation was also observed in the absence of the CDK1 inhibitor RO-3306 (Fig. 6K), which was reported to cause S phase replication to extend into G2/M in some conditions³⁹, indicating that the observed EdU foci reflect MiDAS. In summary, our data suggest that SLX4 engages into a complex with TopBP1 in mitosis via a mechanism that requires both the CDK1-target residue, T1260, and the flanking SIMs of SLX4. Furthermore, SUMOylation seems to be required for promoting MiDAS and the association of SLX4 with TopBP1.

The TopBP1-SLX4 colocalization is important for genome stability in mitosis

Our data suggest that K704 of TopBP1 is required for promoting the recruitment of SLX4 to TopBP1 foci in mitosis. Therefore, we wanted to determine if reduced recruitment of SLX4 to TopBP1 foci in mitosis impacts genome stability. To do so, we examined cells expressing WT or K704A mutant TopBP1 cDNA for MiDAS and formation of 53BP1 nuclear bodies in G₁ following mild replication stress (Fig. 7A). Interestingly, cells expressing the K704A mutant TopBP1 showed a 50% reduction of EdU positive mitotic cells compared to cells expressing WT TopBP1 (Fig. 7B). Similarly, cells deficient in TopBP1-SLX4 colocalization due to expression of



Δ SIM1-3 or T1260A mutant SLX4 also showed marked reduction in MiDAS-positive cells (Fig. 7C). This suggests that the interaction between TopBP1 and SLX4 is important for MiDAS. Consistent, knockdown of endogenous TopBP1 and SLX4 using siRNA, similarly, results in an approximately 50% reduction in EdU positive mitotic cells (Supplementary Fig. 6). Accordingly, we also observed a significant increase in the number of 53BP1 nuclear bodies in the following G₁ of cells expressing the TopBP1-K704A or SLX4-T1260A mutants (Fig. 7D–G), thus showing that cells with reduced TopBP1-SLX4 colocalization transmit damaged DNA to daughter cells.

Discussion

In this study, we have investigated the function of TopBP1 in mitosis after replication stress. Specifically, we provide new insights into the molecular events that follow TopBP1 recruitment to foci in mitosis, which leads to activation of mitotic DNA repair synthesis to promote genome stability after replication stress.

In prophase following replication stress, TopBP1 forms twin foci that colocalize with FANCD2, SLX4, and EdU incorporation (MiDAS) and we observed TopBP1 located between FANCD2 foci as previously reported for MUS81^{14,15}. The colocalization of TopBP1 with FANCD2 foci indicates that

Fig. 6 | SLX4 T1260 and SIMs regulate colocalization with TopBP1 in mitosis.

A Illustration of human SLX4 domain organization with motifs and important residues indicated. Δ SIM mutations are compared to WT sequence. **B** Representative images of mitotic HeLa cells expressing Halo-tagged TopBP1 from the endogenous locus, TFP-tagged H2B and mCherry-tagged WT or mutant SLX4 cDNA. Cells were treated with APH (0.4 μ M) for 16 h before imaging. Endogenous SLX4 was depleted before APH treatment, as described in Fig. 5B. White lines crossing TopBP1-SLX4 foci indicate the line used for quantification of TopBP1-SLX4 intensity values (gray value) plotted in (C). **C** Plots showing the intensity values of TopBP1 (WT) and SLX4 (WT, T1260A, Δ SIM1-2 or Δ SIM1-3) fluorophores along the white lines indicated in (B). **D** Quantification of the percentage of cells with colocalization of at least one SLX4 (WT or mutant) focus with the TopBP1 foci. Colored bullets indicate data from three independent experiments. Mean and error bars indicating 95% confidence-intervals are shown (Student's *t*-test). **E** Quantification of the percentage of cells with colocalization of at least one SLX4 focus with the TopBP1 foci in cells expressing either SLX4-WT or SLX4-T1260A peptides. Colored bullets indicate data from three independent experiments with 3 different clones. Mean and error bars indicating 95% confidence-intervals are shown (Student's *t*-test). Statistical test based on Student's *t*-test ($n = 72$ (SLX4-WT) or $n = 66$ (SLX4-T1260A)). *, $p < 0.05$. **F** Quantification of fluorescence intensities of colocalizing TopBP1 and SLX4 foci in cells expressing either SLX4-WT or SLX4-T1260A peptides. Plotted values indicate the intensity-ratio between TopBP1 and SLX4 at colocalizing foci, normalized to the total SLX4 signal per nucleus. Data is plotted as super-plot in three colors, indicating data points from three independent experiments with three different clones. Mean and error bars indicating 95% confidence intervals are shown. Statistical test based on Student's *t*-test ($n = 327$ (SLX4-WT) or $n = 299$ (SLX4-T1260A)). *****, $p < 0.0001$. **G** Quantification of nuclear

SUMO2/3 staining intensity in control condition or after treatment with TAK-981 at indicated concentrations. Colored bullets indicate data from three independent experiments. >100 cells per condition were quantified for each replica by automated image analysis. Mean and error bars indicating 95% confidence-intervals are shown. Statistics were evaluated by Student's *t*-test. ***, $p < 0.001$. *****, $p < 0.0001$.

H Representative images of mitotic cells expressing Halo-tagged TopBP1 from the endogenous locus, TFP-tagged H2B and mCherry-tagged WT SLX4 cDNA in the absence or presence of TAK-981 (4 h treatment). Endogenous SLX4 was depleted prior to treatment, as described in Fig. 5B. White lines crossing TopBP1-SLX4 foci indicate the line used for quantification of TopBP1-SLX4 intensity values (gray value) plotted in (I). White arrows indicate the two foci, which are intersected by the white line. **I** Plot showing the intensity values of TopBP1 (WT) and SLX4 (WT) along the white lines indicated in (H). Dashed black line indicates the background intensity of SLX4. **J** Quantification of fluorescence intensities of colocalizing TopBP1 and SLX4 foci in the absence or presence of TAK-981 (quantified from data in (H)). Plotted values indicate the intensity-ratio between TopBP1 and SLX4 at colocalizing foci, normalized to the total SLX4 signal per nucleus. Data is plotted as super-plot in two colors, indicating data points from the two independent experiments. Means are indicated by horizontal lines ($n = 131$ (Control) or $n = 103$ (TAK-981)).

K Quantification of total EdU foci per metaphase spread with low dose APH (0.4 μ M, 16 h) and with or without SUMO-inhibition by TAK-981 (1 μ M, 2 h). Analysis performed on cells synchronized with CDK1 inhibitor RO-3306 or without RO-3306. Mean and error bars indicating 95% confidence intervals are shown. Statistical test based on Student's *t*-test (RO-3306, $n = 284$ (DMSO) or $n = 292$ (TAK-981); No RO-3306, $n = 63$ (DMSO) or $n = 63$ (TAK-981)). *****, $p < 0.0001$. **B–J** CDK1-inhibition was not used in these experiments.

the incorporated EdU stems from a locus where replication was stalled before MiDAS was initiated. SLX4 recruitment and EdU incorporation are largely dependent on TopBP1, which is consistent with SLX4-associated nucleases catalyzing incisions that allow break-induced replication to start^{5,9}. However, it is noteworthy that we do observe some residual EdU foci that colocalize with FANCD2 after depletion of TopBP1 (Fig. 1D). This indicates that alternative TopBP1-independent pathways may exist to facilitate DNA synthesis in mitosis or that enough TopBP1 remains to initiate some MiDAS in our experimental conditions. An alternative explanation could be that some stalled forks collapse, leading to extensive ssDNA that may serve as a substrate for RAD51 in a TopBP1-independent salvage pathway, where the incision performed by the SLX4-MUS81 complex may not be necessary to activate MiDAS. In support of this scenario, RAD51 was recently reported to promote MiDAS by protection of the ssDNA at non-DDR associated loci in a process that does not involve RAD52¹⁷. Remarkably, while FANCD2 and EdU generally remain symmetrically located to twin foci on sister chromatids until metaphase, one of the TopBP1 twin foci is often disassembled in a manner dependent on DNA synthesis and the remaining TopBP1 focus is associated with a chromatin gap. Based on these findings, we propose the following model for the role of TopBP1 in promoting MiDAS. In S-phase, FANCD2 is recruited to stalled replication forks, which leads to formation of the characteristic twin foci at underreplicated regions⁴⁰. Upon mitotic entry, TopBP1 forms twin foci that co-localize with FANCD2 twin foci. Next, TopBP1 promotes the recruitment of SLX4, which in turn recruits MUS81 to cleave the stalled forks to produce one-ended DSBs at either end of the underreplicated region. Hereafter, MiDAS is promoted by RAD52-mediated annealing of the resected one-ended DSB to the complementary strand on the sister chromatid. Annealing is followed by POLD3-dependent DNA synthesis. MiDAS shares many of the genetic requirements known from BIR⁸. Moreover, we find that EdU often presents as twin foci on metaphase chromosomes, suggesting that MiDAS is likely to be initiated at both of the converging stalled forks, given that MiDAS occurs by conservative replication⁵. Interestingly, we find that the remaining TopBP1 foci in late mitosis are almost exclusively found at chromatin gaps and ends that coincide with FANCD2 and EdU foci. Upon completion of DNA repair synthesis, TopBP1 dissociates from chromatin. Thus, the remaining TopBP1 at FANCD2-EdU positive chromatin gaps could reflect incomplete replication as previously indicated by the transition of the

remaining TopBP1 foci in anaphase into 53BP1 nuclear bodies in the following G1²⁰.

As noted above, TopBP1 was recently assigned a role in mitotic DSB repair, where it tethers broken chromosome ends generated by ionizing radiation in an MDC1-dependent manner^{41,42}. In addition to this role, we find that nearly all (>95%) replication stress-induced TopBP1 foci that localize to chromatin gaps also colocalize with EdU (Fig. 4F), which suggests that in response to replication stress the main role of TopBP1 in mitosis is to promote MiDAS rather than tethering of broken chromosomes. Supporting this notion, another recent study reported MDC1-independent recruitment of TopBP1 to foci in mitosis following mild replication stress⁴³.

We dissected the recruitment of SLX4 to TopBP1 foci in mitosis. Interestingly, we found that K704 within BRCT5 of TopBP1 is important for recruitment of SLX4 to replication stress induced foci in mitosis. Additionally, we also investigated how SLX4 contributes to the interaction. Interestingly, we found that the TopBP1-SLX4 interaction functions in a dual-mode, which is both dependent on the docking of SLX4, likely through phosphorylated T1260 onto K704 of TopBP1, and on the SUMO interaction motifs flanking T1260 of SLX4. Inhibition of SUMO-conjugation also strongly reduced the recruitment of SLX4 to TopBP1 foci and MiDAS, supporting a function of the SIMs in recruitment of SLX4 to sites of underreplicated DNA in mitosis to initiate MiDAS. Interestingly, a recent study showed that condensation of SLX4 into foci depends on its three SIM motifs⁴⁴, suggesting that liquid-liquid phase separation of SLX4 also contributes to its recruitment to TopBP1 foci. Notably, TopBP1 foci are largely unaffected by mutation of the SIMs in SLX4, indicating that the reported liquid-liquid phase separation of TopBP1⁴⁵ is controlled by a different mechanism. The sumoylation sites bound by SLX4 SIMs are not known, but notably TopBP1 is one of the most highly sumoylated proteins in response to replication stress³⁴. Alternatively, SLX4 itself is also sumoylated in mitosis and has been shown to act as an E3 SUMO ligase for its own sumoylation^{35,46}.

Given that BLM, BRCA1, 53BP1, and Treacle also interact with the BRCT5 domain of TopBP1⁴⁷, it remains an open question for future studies to which extent these interactions compete with the SLX4 interaction. With the number of molecules of each of these proteins per cell estimated at 200 (SLX4), 10,211 (TopBP1), 26,053 (BLM), 11,985 (BRCA1), 127,144 (53BP1), and 55,576 (Treacle)⁴⁸, there is clearly potential for competition in binding of these proteins to TopBP1. It is likely that timing of

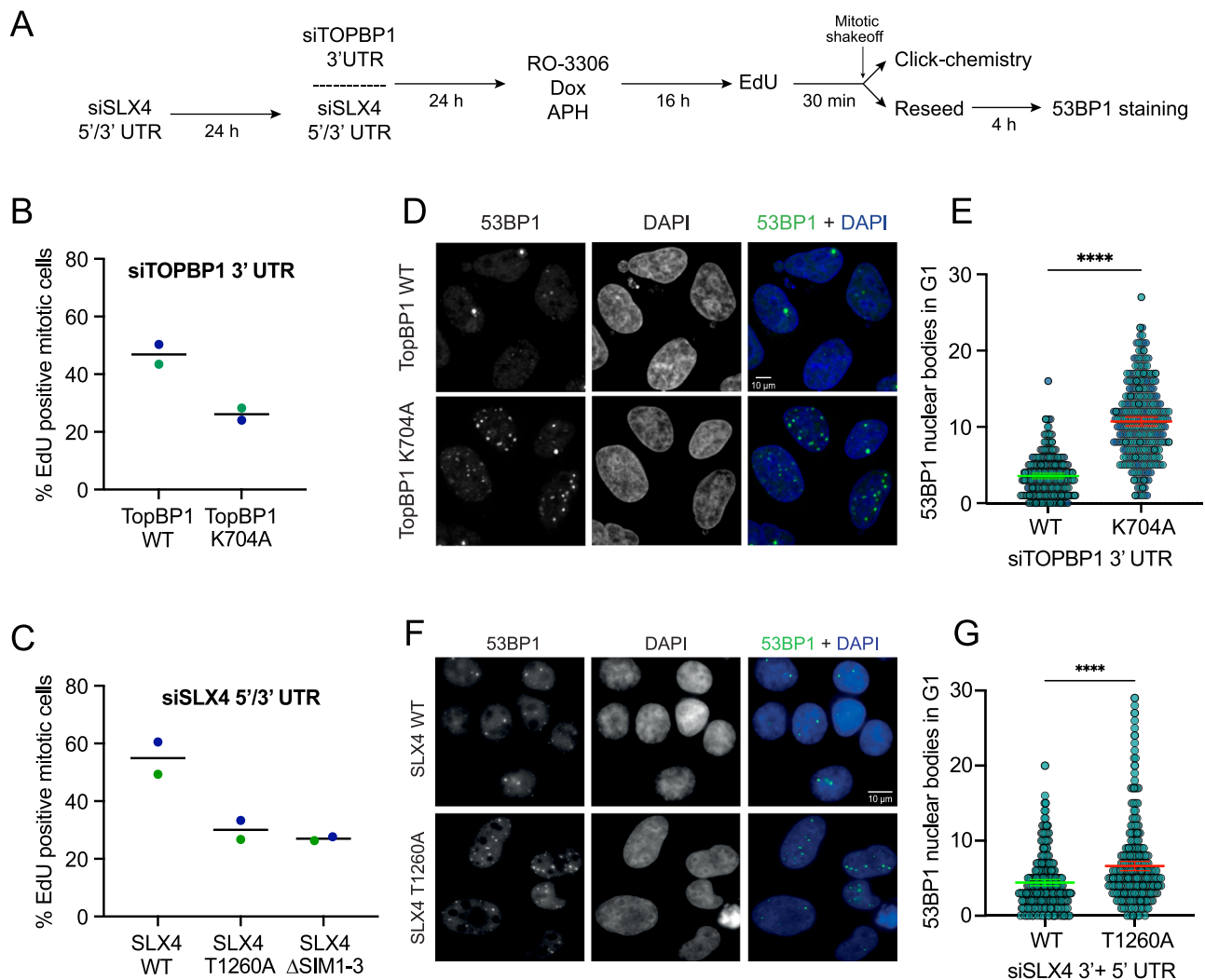


Fig. 7 | TopBP1 K704, SLX4 SIMs and T1260 promote genome stability in mitosis. **A** Experimental scheme. Mitotic cells were reseeded in suspension into an 8-well chambered Ibidi slide in DMEM (10% FBS, 1% P/S), followed by 4 h incubation for the cells to adhere to the bottom. Hereafter, cells were fixed for 10 min in 4% formaldehyde and stained for 53BP1. **B, C** Quantification of mitotic cells with EdU foci (in percentage) in cells expressing WT or mutant variants of TopBP1 or SLX4. Expression of cDNA was induced by DOX after knockdown of endogenous TopBP1 or SLX4 using siRNA targeting the 3' UTR or 5' and 3' UTR, respectively (see Western blots in Supplementary Fig. 7). Expression of WT or mutant TopBP1 or SLX4 cDNA from an FRT-site were performed as outlined in (A) ($n = 2$). > 300 cells were quantified per condition across the two replicates. CDK1i (RO-3306) was used for G2-synchronization prior to mitotic release. Means are indicated by horizontal

lines. **D** Representative images of immunofluorescence staining for 53BP1 foci in G₁ cells after transition through mitosis with TopBP1 K704A as outlined in (A). DAPI was used as counterstain. **E** Quantification of 53BP1 nuclear bodies per G₁ cell in images from (D). Mean and error bars indicating 95% confidence intervals are shown. Statistical test based on Student's *t*-test ($n = 287$ (WT) or 284 (K704A)). ****, $p < 0.0001$. **F** Representative images of immunofluorescence staining for 53BP1 foci in G₁ cells after transition through mitosis with SLX4 T1260A as outlined in (A). DAPI was used as counterstain. **G** Quantification of 53BP1 nuclear bodies per G₁ cell in images from (F). Mean and error bars indicating 95% confidence intervals are shown. Statistical test based on Student's *t*-test ($n = 287$ (WT) or 278 (T1260A)). ****, $p < 0.0001$.

phosphorylation of the different interaction partners controls their interaction with TopBP1.

In summary, we find that TopBP1 promotes MiDAS by recruiting SLX4 in mitosis. The recruitment is likely mediated through a phosphorylation-dependent interaction between T1260 of SLX4 and K704 of TopBP1. The interaction is further enhanced in a SUMOylation dependent manner, which relies on the SUMO interaction motifs that flank the proposed TopBP1-binding site of SLX4. This interaction promotes genome stability in mitosis following replication stress, as reduced interaction results in decreased MiDAS and elevated levels of 53BP1 nuclear bodies in the next G₁. Based on our findings, we propose that inhibiting the TopBP1-SLX4 interaction may constitute a candidate target for cancer therapy in cancers that rely on MiDAS to relieve replication stress caused by oncogene activation.

Methods

Cell lines and cell culture conditions

Cell lines used in this study are listed in Supplementary Table 1. All cell lines used were cultured in Dulbecco's Modified Eagle Medium (DMEM), high glucose, GlutaMAX™ supplemented with 10% FBS and 1% Penicillin/Streptavidin (10,000 units/mL of penicillin and 10,000 µg/mL of streptomycin, Gibco by Life Technologies, cat. no. 15140-122) and grown at 37 °C with 5% CO₂. Additionally, Flp-in cell lines were cultured in DMEM supplemented with 5 µg/mL Blastidin and 200 µg/mL Hygromycin. Cell lines tested negative for mycoplasma contamination. None of the cell lines used were specifically authenticated for this study. HeLa cells were used in this study because of the suitability for microscopy and because they are widely used in the DNA repair field. All cell lines are shared upon request to the corresponding author.

Construction of plasmids

Plasmids used in this study are listed in Supplementary Table 2 and primers are listed in Supplementary Table 3. For Dox-inducible expression of mCherry-tagged cDNA, plasmid pSNK011 was created by subcloning of mCherry from pmCherry-C1 into AflIII/KpnI-digested pTMN7. Plasmid pSNK013 was created by subcloning of TopBP1 WT cDNA from pEGFP-C1-hTopBP1 into KpnI/NotI-digested pSNK011. pSNK014 was subsequently created by site-directed mutagenesis of K704 of TopBP1 in pSNK013 using QuikChange (Life Technologies) and primers 5' FW K704A and 3' RV K704A. For pJB064, SLX4 cDNA was amplified from pJB017 and adapted with 5' KpnI and a start codon and 3' KpnI overhangs. SLX4 cDNA was inserted into the KpnI site of pSNK011. pJB065 was subsequently created by site-directed mutagenesis of T1260 of SLX4 in pJB064 using QuikChange and primers 5' FWT1260A and 3' RV T1260A. For pJB066, Δ SIM1-2 fragment cDNA was ordered from GeneArt (Life Technologies) and inserted into the HindIII site of pJB064. For pJB071, Δ SIM3 fragment cDNA was ordered from GeneArt (Life Technologies) and inserted into pJB066, using XmaI, thus yielding a Δ SIM1-3 mutant cDNA.

For transient expression of mVenus-tagged SLX4 cDNA, SLX4 WT or T1260A cDNA was isolated from pJB064 or pJB065, respectively, using KpnI digestion and inserted into pJB072, yielding pJB067 and pJB068, respectively.

Vectors pJJ2 and pJJ3 expressing the WT SLX4 peptide SWLVPA-PLASRSRDCS or the SLX4-T1260A mutant peptide SWLVPA-PLASRSRDCS, respectively, fused to a Destabilization Domain (DD)³⁷, were constructed by PCR-mediated extension of DD with the SLX4 peptide-coding sequence using primers DD-HindIII-FW and DD-Tpep-RV/DD-Apep-RV and pBMN-FKBP(DD)-YFP as a template. Next, the PCR product was digested with HindIII/NotI and cloned into HindIII/NotI-digested pEGFP-N1-TetR downstream of the CMV promoter. All plasmids are shared upon request to the corresponding author.

Construction of cell lines

We used a CRISPR-Cas9 genome editing approach to endogenously tag TopBP1 on both alleles with the Halo-tag. In short, Cas9-D10A-mediated nicking on both strands flanking the endogenous STOP-codon allowed insertion of the Halo-tag at the C-terminus of TopBP1 by homology-directed repair in cells co-transfected with a repair-template containing the halo-tag, a removable Puromycin, Neomycin or Blasticidin resistance cassette and 5'/3' homology arms (pJB007, pJB008 and pJB005 or pJB049 or pTMN4). This procedure was performed for all TopBP1-halo tagged cell lines.

For random integration of H2B-TFP into a HeLa T-REX cell line (background for dox-inducible expression of mCherry-tagged cDNA), WT HeLa T-REX cells were transfected with pJB030 using Lipofectamine 3000 (according to manufacturer's protocol). Hereafter, cells stably expressing the TFP-tagged H2B cDNA were selected using 500 μ g/ml G418 for 2 weeks. Media was changed regularly.

To stably integrate pcDNA5 plasmid variants into the FRT-site, HeLa T-REX H2B-TFP cells were transfected with pcDNA5/FRT/TO-mCherry-cDNA plasmids and pOG44, encoding the Flp-recombinase, in a 1:9 ratio using Lipofectamine 3000 (according to manufacturer's protocol). Hereafter, cells stably expressing Hygromycin resistance cassette were selected with 200 μ g/ml hygromycin. Media was changed regularly.

To generate cells expressing SLX4 T1260/T1260A peptides, HeLa T-REX H2B-TFP cells with pcDNA5/FRT/TO-mCherry-SLX4-WT and TopBP1-Halo/+ were transfected with BsaI-linearized plasmids pJJ2 or pJJ3 (expressing SLX4 T1260/T1260A peptides fused to an FKBP12 destabilizing domain) using Lipofectamine 3000 for random integration. Hereafter, cells stably expressing G418 resistance cassette were selected with 500 μ g/ml G418. Media was changed regularly.

To generate the TopBP1-mAID-SMASH cell line a CRISPR-Cas9 approach was used, similar to the approach for endogenous tagging of TopBP1 with Halo-tag (described above). Of note, the RPE1 background

cells were engineered with a dox-inducible expression system for OsTIR with a tetracycline responsive TRE promoter, located at the *rosa26* safe-harbor locus⁴⁹.

Transfection of siRNAs and plasmids

For plasmid transfection, cells were electroporated using the NEON transfection system using manufacturer's protocol (Life Technologies). In short, a cell suspension with a density of $5.0 \cdot 10^6$ cells/ml was prepared in resuspension buffer (buffer R). Next cells were mixed with 250 ng of plasmid and submitted to electroporation with parameter adjusted to HeLa cells, according to manufacturer's protocol.

For siRNA transfection, cells were reverse transfected with 20 nM (siSLX4 5'/3' UTR) or 40 nM (siTopBP1 3' UTR) of siRNA, using Lipofectamine RNAiMAX (Thermo Fisher Scientific, 13778075). For knock-down of endogenous SLX4, a second round of siRNA transfection was performed as forward transfection on the plated cells with the same transfection protocol. siRNA used in this study are listed in Supplementary Table 4.

SDS-Page and Western blotting

For TopBP1 and SLX4, protein lysates were separated on a 4–20% gradient Tris/glycine SDS-PAGE gel (Bio-Rad) and transferred to a 0.45 μ m nitrocellulose membrane overnight at 30 V at 4 °C. Subsequently, the membrane was blocked (5% skimmed milk in TBS with 0.1% Tween 20 (TBS-T)) for 1 h at room temperature. After blocking, the membrane was incubated with primary antibodies against TopBP1 (1:2000, A300-111A, Bethyl Laboratories), SLX4 (1:1000, gift from John Rouse, University of Dundee), and Tubulin (1:2000, Abcam, ab6160) for 1 h at room temperature. For FKBP12, protein lysates were separated on 4–12% NuPAGE Bis-Tris gels (Invitrogen) and transferred to a 0.22 μ m nitrocellulose membrane 1 h at 80 V. The membrane was blocked and incubated with primary antibody against FKBP12 (1:500, sc-133067, Santa Cruz) overnight at 4 °C. Subsequently, the membrane was washed three times with TBS-T for 5 min and probed with HRP-conjugated secondary antibodies for 1 h at room temperature. Western blot was visualized using ECL (Cytiva, RPN2134).

TopBP1-degradation assays

RPE1 cells (TopBP1^{mAID-SMASH}/mAID-SMASH⁺, OsTIR) were seeded in a T75 cell culture flask at approximately 30% confluency and left to adhere in the incubator. On the next day, cells were treated for 16 h with RO-3306 (10 μ M) to arrest cells at the G₂/M border in the presence of a low dose of aphidicolin (0.5 μ M) to induce mild replication stress. Additionally, cells were treated with doxycycline (1 μ g/ml) to induce expression of OsTIR, which is required for AID-mediated protein degradation of TopBP1-mAID-SMASH. TopBP1-positive samples (no degradation control) were left in the incubator for 16 h. Alternatively, samples subject to TopBP1 degradation were kept in the incubator for 10 h and subsequently treated with Auxin (500 μ M, for AID-tag degradation) and the ASV protease-inhibitor (3 μ M, for SMASH-tag degradation, Insight Biotechnology Limited, HY-14434-5MG) for 6 h. Hereafter, cells were released into mitosis by two washes of pre-heated DMEM medium. Next, cells were pulsed with EdU (20 μ M), for 35 min at 37 °C as described⁹. Mitotic cells were isolated by mitotic shake-off and spun onto glass slides using a Cytospin centrifuge (Thermo Fisher Scientific) for 5 min at 1000 rpm. Samples were fixed with 3% paraformaldehyde (in PBS, pH 7.4, freshly made) for 10 min at room temperature. Fixation was followed by three washes in ice-cold PBS containing 3% BSA. Hereafter, samples were permeabilized for 20 min at room temperature with 0.1% Triton X-100 (in PBS), followed by two washes in PBS containing 3% BSA. EdU was detected using the A647-EdU detection kit and performed according to manufacturer's protocol (Thermo Fisher Scientific, C10340). For immunofluorescence detection, samples were incubated with mouse anti-mAID (1:200, MBL, M214-3) and rabbit anti-FANCD2 (1:500, Novus, NB100-182) in a humidifying chamber at 37 °C for 30 min. Hereafter, samples were washed in PBS-T and incubated with anti-mouse-A488 (1:1000, Thermo Fisher Scientific, A11029) and anti-rabbit-

A594 (1:500, Thermo Fisher Scientific, A32740) in a humidifying chamber at 37 °C for 30 min. Lastly, samples were sealed and DNA counterstained with mounting buffer containing DAPI (Thermo Fisher Scientific, D1306).

For degradation of Halo-tagged TopBP1, 500 nM HaloProtac-E (MRC PPU Reagents and Services, University of Dundee) was added to the cell culture media for 16 h²⁴.

Halo-labeling procedure

For live cell Halo-labeling, a 5× TMR Halo-ligand working solution was prepared according to manufacturer's protocol in pre-heated culture medium (DMEM + 10%FBS + 1%P/S). In a well containing Halo-tagged cells, one fifth of the culture medium was replaced with the same amount of TMR working solution (5×) and mixed carefully to avoid cell detachment. The cells were labelled with the TMR ligand for 15 min in an incubator (37 °C, 5% CO₂). Hereafter, cells were washed twice with medium and incubated in fresh, pre-heated medium for another 30 min in the incubator to wash away unbound ligand. Same approach was used to label TopBP1 with Janelia 646 (JF646, Promega, cat.no. GA1120 Halo-ligand).

Preparation of native metaphase spreads

Two days before harvest, cells were seeded in a T75 flask and incubated overnight (37 °C, 5% CO₂). The next day, cells were left untreated or treated with APH (0.4 μM) and RO-3306 (10 μM) for 16 h. Prior to G2-release and EdU pulse, cells were washed twice with pre-warmed culture medium. Mitotic synchronization was achieved with 35 min to 1 h of Colcemid treatment (100 ng/ml, Thermo Fisher Scientific, 15212012) in the incubator post G2 release. Together with Colcemid treatment, cells were EdU-pulsed with 20 μM EdU. Following EdU pulse/metaphase enrichment, cells were harvested by mitotic shake-off and samples were centrifuged for 2 min at 300 g. Labelling of TopBP1-Halo with fluorescent halo-ligand was performed as described above except for skipping the 30 min wash step. Thus, after 15 min of Halo-labelling, samples were resuspended in hypotonic buffer and native metaphase spreads were prepared according to ref. 50 with minor modifications. Pelleted cells were resuspended in fresh 2 ml ice-cold hypotonic buffer (1:5 FBS, 1:5 75 mM KCl, 3:5 ddH₂O) while vortexing at low speed. Hereafter, samples were spun onto microscope slides using the Cytospin centrifuge (Thermo Fisher Scientific) for 5 min at 1000 rpm. Cells were fixed with 3% paraformaldehyde (in PBS, pH 7.4) for 10 min at room temperature. Next, slides were immersed into Coplin jars containing KCM buffer (120 mM KCl, 20 mM NaCl, 10 mM Tris-HCl pH 8.0, 0.5 mM EDTA, 0.1% Triton X-100, Thermo Fisher Scientific, T8787). After permeabilization, slides were washed twice with PBS for 5 min in the Coplin jar. Slides were blocked in blocking solution (3% BSA (w/v) in PBS containing 0.1% Tween 20 (PBS-T)) for 30 min at room temperature. After blocking, slides were washed twice in a Coplin jar with PBS-T for 3 min. Subsequently, samples were incubated with primary antibodies (in blocking solution), followed by two PBS-T washing steps in a Coplin jar and lastly incubated with fluorophore-conjugated secondary antibodies (in blocking solution). All antibody incubation steps were performed in a humidifying chamber at 37 °C for 30 min. Primary antibodies used: Mouse anti-GFP (1:1000, Roche, 11814460001), Rabbit anti-Halo (1:500, Promega, G9281), Sheep anti-SLX4 (1:100, S714C, gift from John Rouse, University of Dundee), Rabbit anti-FANCD2 (1:500, Novus, NB100-182SS). Secondary antibodies used: Goat anti-Mouse-Alexa594 (1:500, Thermo Fisher Scientific, A11005), Goat anti-Mouse-STAR580 (1:500, Abberior, ST580-1001), Goat anti-Mouse-Alexa488 (1:500, Invitrogen, A11029), Goat anti-Rabbit-STAR RED (1:500, Abberior, STRED-1002), Goat anti-Rabbit-Alexa647 (1:500, Invitrogen, A21245), and Donkey anti-Sheep-Alexa350 (1:500, Invitrogen, A21097). After antibody probing, samples were mounted with Prolong Diamond (Thermo Fisher Scientific, P36961) and allowed to cure for 24 h. For visualization of EdU, click-reaction was performed using the EdU click-chemistry kit with fluorescent azides (Az-405 (#1477) and Az-532 (#1476), clickchemistrytools) according to manufacturer's protocol.

TopBP1-FANCD2 asymmetry in early mitosis

To assess the relative localization of TopBP1 and FANCD2 twin foci in early mitosis, cells (NCAPH-mEGFP TopBP1-Halo) were seeded in an 8-well μ-slide (Ibidi, cat # 80826) and allowed to adhere overnight. Sixteen hours before fixation, cells were treated with a low dose of aphidicolin (0.4 μM) to induce mild replication stress. One hour before fixation, TopBP1 was labelled with the TMR halo-ligand (Promega, cat# G8251) according to manufacturer's protocol. Cells were fixed for 10 min in 4% formaldehyde and permeabilized for 5 min in 0.5% Triton X-100 at room temperature. Hereafter, samples were blocked using 3% BSA (in PBS-T) for one hour at room temperature. Following blocking, samples were stained with a primary antibody against FANCD2 (Novus Biologicals, NB100-182SS, 1:500) for 1 h at 37 °C. After primary antibody incubation, samples were washed 3 × 5 min with PBS-T (0.1%), followed by incubation with a secondary antibody conjugated to STARRED (Abberior, STRED-1002, 1:500) for 1 h at room temperature. Finally, samples were counterstained with DAPI (200 ng/ml) for 10 min at room temperature and washed once in auto-claved mQH₂O.

TopBP1-SLX4 colocalization in early mitosis

To investigate colocalization of TopBP1 and SLX4 in prometaphase cells, cells were seeded in an 8-well μ-slide (Ibidi, cat # 80826) and grown for 48 h. Cells were then treated with 0.4 μM APH, 1 μg/mL doxycycline for mCherry-SLX4-WT expression, and 1 μM Shield-1 (Abmole, M9704) for 16 h. Cells were labelled with Halo-ligand JF646 for 15 min as previously described. Before live-cell imaging cells were incubated for 30 min (37 °C, 5% CO₂). TopBP1 and SLX4 foci intensities were quantified and data is presented as the percentage of cells with TopBP1-SLX4 colocalization or as SLX4/TopBP1 intensity ratio normalized to the total amount of nuclear SLX4.

Fluorescence microscopy

Fluorescence microscopy was performed using a wide-field microscope (DeltaVision Elite; Applied Precision) equipped with a 100× objective lens with a numerical aperture of 1.4 (U-PLAN S-APO, NA 1.4; Olympus), a cooled EMCCD camera (Evolve 512; Photometrics), and a solid-state illumination source (Insight; Applied Precision, Inc.). Immunofluorescence images were taken at room temperature with 10 optical sections separated by 0.33 μm. Live cell images were acquired at 37 °C. Images were acquired with SoftWoRx 7.0.0 (Applied Precision/GE Healthcare) software. Processing and quantitative measurements of fluorescence intensities were performed with Volocity 6.3 software (PerkinElmer).

TAK-981 SUMO-inhibition assay

Cells were seeded in a 96-well CellCarrier Ultra plate (PerkinElmer) and incubated overnight. Next day, DMSO (control) or increasing concentrations of TAK-981 (0.01 μM–1 μM, Selleck Chemicals LLC, S8829) was added to the cells and incubated for 4 h. Hereafter, cells were fixed with 3% paraformaldehyde for 10 min at room temperature. Next, cells were permeabilized with 0.5% Triton X-100 (in PBS) for 5 min at room temperature. After permeabilization, cells were washed twice with PBS and blocked with 3% BSA (in PBS-T) for 1 h at room temperature. Following blocking, mouse anti-SUMO2/3 (1:1000, MBL, M114-3) in blocking solution was added to the cells. Antibody incubation was performed for 1 hour at room temperature. The cells were then washed twice with PBS-T for 3 min and incubated with anti-mouse-Alexa561 (1:500, Invitrogen, A11031) for 1 h at room temperature. Lastly, cells were washed twice with PBS. Imaging was performed using a HCS confocal microscope (OPERA-QEHS, PerkinElmer) and automated image analysis was performed using the Columbus server (PerkinElmer).

Statistics and reproducibility

Statistical significance was evaluated with GraphPad Prism 10.4.0 using two-sided Student's *t*-test as indicated in figure legends. For microscopy and western blots, representative images are shown. Unless otherwise stated 3

biological replicates were performed. A biological replicate is defined as an independent experiment with new biological material.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All other data underlying this article will be shared on reasonable request to the corresponding author. Uncropped Western blots are shown in Supplementary Figs. 8 and 9.

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Author contributions

M.L., J.B., and V.H.O. conceived, designed, and supervised the study. Experiments were performed by J.B., K.V.P., S.N.K., T.M.N., J.R., C.R.B.,

and J.J. The human cell line expressing mAID–S-tagged TopBP1 was constructed and characterized by M.S. and A.C. I.A.H. provided analysis of sumoylation sites. The manuscript was written by J.B., and all authors reviewed, commented and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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